Blockchain Security: The Threat from Quantum Computing is Real

Blockchains must gear up to protect themselves from quantum computing-based attacks. Post-quantum cryptography and artificial intelligence offer some solutions.

Blockchain is a shared and distributed digital database that holds transactions in an immutable and secure way, without relying on any intermediary. It employs cryptographic mechanisms such as hashing and public-key cryptography to preserve the information unaltered and resistant to unauthorised tampering. Blockchain technology has revolutionised the digital economy by using a secure, decentralised way of recording transactions without an intermediary. However, the rapid progress of quantum computing has the potential to tear down the security of the blockchain.

Traditional blockchain security depends on cryptography such as RSA and ECC (elliptic curve cryptography), which ensures data integrity and authenticity against unauthorised access. Such cryptography is immune to conventional computers but susceptible to quantum algorithms that have the ability to decrypt cryptography by factoring massive prime numbers much quicker than current supercomputers. This implies that once quantum computers come into regular use, they can decrypt the blockchain encryption, revealing transactions, digital coins, and user identities.

Web3, the emerging next-generation web, is constructed on the potential of blockchain by unveiling decentralised applications (dApps) and smart contracts, which offer a trustless and permissionless environment. The security of blockchain networks is based on cryptographic techniques like the Elliptic Curve Digital Signature